

Loan Timing Around the End of a Loan

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Overview

This week I looked more closely at what happens around the end of a loan, with two main questions:

1. How long is the “short window” between a player’s loan return and a subsequent permanent transfer back to the borrowing club?
2. Are loan endings concentrated on a small number of standard calendar dates, such as 31 May, 30 June, or 31 December?

The main timing analysis focuses on the strict sequence

$A \rightarrow B$ loan

$B \rightarrow A$ End of loan

$A \rightarrow B$ permanent transfer.

There are **4,068 exact instances** of this sequence in the current dataset. In every case, the same player moves from the exact lender club A to borrower B on loan, returns exactly from B to A , and then moves permanently from A back to B . There are no intervening transfer-history rows between the return and the permanent move.

The timing variable used below is

lag days = permanent-transfer date – loan-return date.

The dates are the dates recorded by Transfermarkt. They therefore describe registration timing, not necessarily the date on which the permanent transfer was negotiated or agreed.

Three related counts

Three counts appear in the analysis and refer to different populations:

Count	Meaning
4,068	Strict $A \rightarrow B, B \rightarrow A, A \rightarrow B$ sequence, at any lag. This is the primary timing population used below.
4,109	Broader set in which the player’s next permanent move goes to borrower B , but the move does not necessarily originate from the exact lender club A . There are only 41 such additional cases. (e.g. Atalanta U19 to Club B. Club B to Atalanta U19. Atalanta to Club B permanent.)
3,818	The earlier production classification of loan-to-permanent cases. This uses the current 60-day follow-on window, so it is a window-dependent production count rather than the population for the timing analysis.

1 How long is the “short window”?

1.1 Distribution of the return-to-permanent lag

Figure 1 shows all 4,068 strict sequences grouped into broad lag intervals. The figure uses the full vertical scale, so the size of the day-1 spike is visible directly.

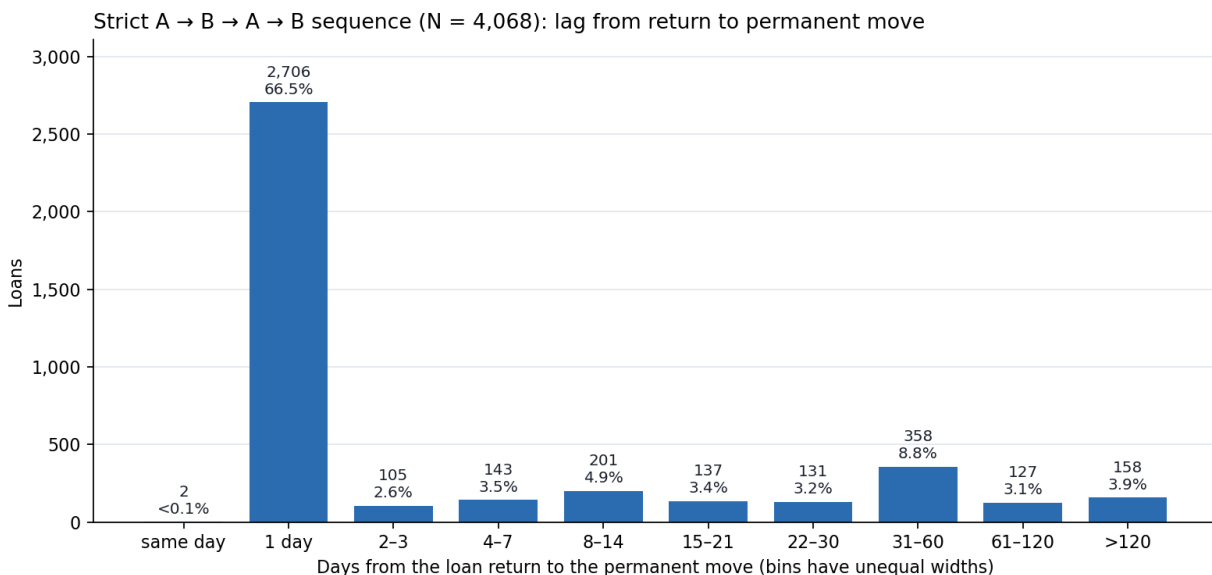


Figure 1: Distribution of the number of days from the loan return to the permanent move back to the same borrower for the 4,068 strict $A \rightarrow B \rightarrow A \rightarrow B$ sequences. The bins have unequal widths, so bar height represents the number of loans in each interval, not density per day.

The exact distribution is:

Lag after return	Loans	Share
Same day	2	< 0.1%
Exactly 1 day	2,706	66.5%
2–3 days	105	2.6%
4–7 days	143	3.5%
8–14 days	201	4.9%
15–21 days	137	3.4%
22–30 days	131	3.2%
31–60 days	358	8.8%
61–120 days	127	3.1%
More than 120 days	158	3.9%
Total	4,068	100.0%

The corresponding cumulative numbers are:

Maximum lag	Loans	Share
Within 1 day	2,708	66.6%
Within 3 days	2,813	69.1%
Within 7 days	2,956	72.7%
Within 21 days	3,294	81.0%
Within 30 days	3,425	84.2%
Within 60 days	3,783	93.0%

The median lag is **1 day**, the 90th percentile is **43 days**, and the 95th percentile is **75 days**.

1.2 Is there a natural cutoff?

The most striking feature is the discontinuity immediately after one day.

There are:

2,706

permanent moves exactly one day after the return, compared with only

50

on day 2.

The busiest later single day is day 31, with 101 cases. Thus, the main empirical discontinuity is between day 1 and day 2. There is no comparable break around either 21 or 60 days.

Figure 2 shows the same result cumulatively.

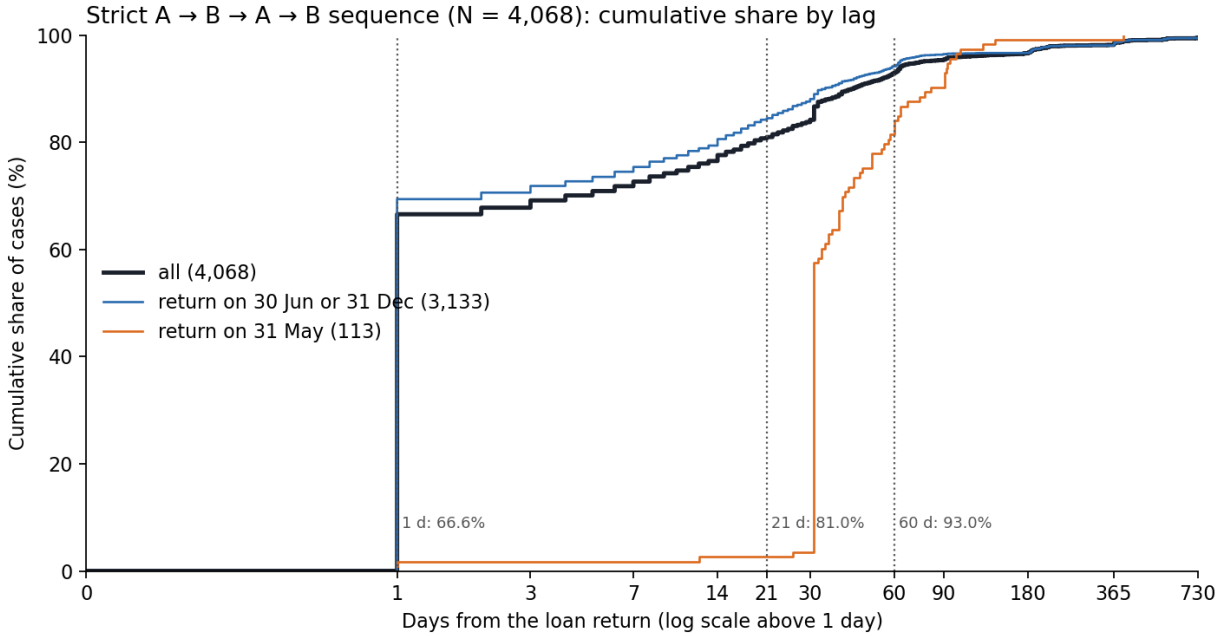


Figure 2: Cumulative share of the 4,068 strict loan-return-permanent sequences by lag. The existing 21- and 60-day production windows are shown for reference, but neither corresponds to a visible break in the distribution.

The current production windows therefore have useful coverage properties:

$$21 \text{ days} \rightarrow 81.0\%$$

and

$$60 \text{ days} \rightarrow 93.0\%,$$

but the data do not identify either threshold as a natural boundary. The clearest empirical cluster is the next-day move.

1.3 Realised-only robustness

Some transfer-history rows were still scheduled future moves when the player's history was captured. To check that these observations are not driving the result, I repeated the calculation requiring both the return and the permanent transfer to have already occurred.

There are **3,936** strict sequences with both dates realised.

Among these:

Statistic	Realised-only result
Within 1 day	2,585 / 3,936 = 65.7%
Within 3 days	68.3%
Within 7 days	72.0%
Within 21 days	80.6%
Within 60 days	92.8%
Median	1 day
p90	44 days
p95	77 days

The timing pattern is therefore essentially unchanged when scheduled moves are excluded.

2 What drives the next-day conversions?

The next question is whether the 2,706 day-1 conversions occur randomly throughout the calendar or are concentrated around standard loan-ending dates.

Figure 3 breaks the 2,706 exact day-1 conversions down by the date on which the loan return was recorded.

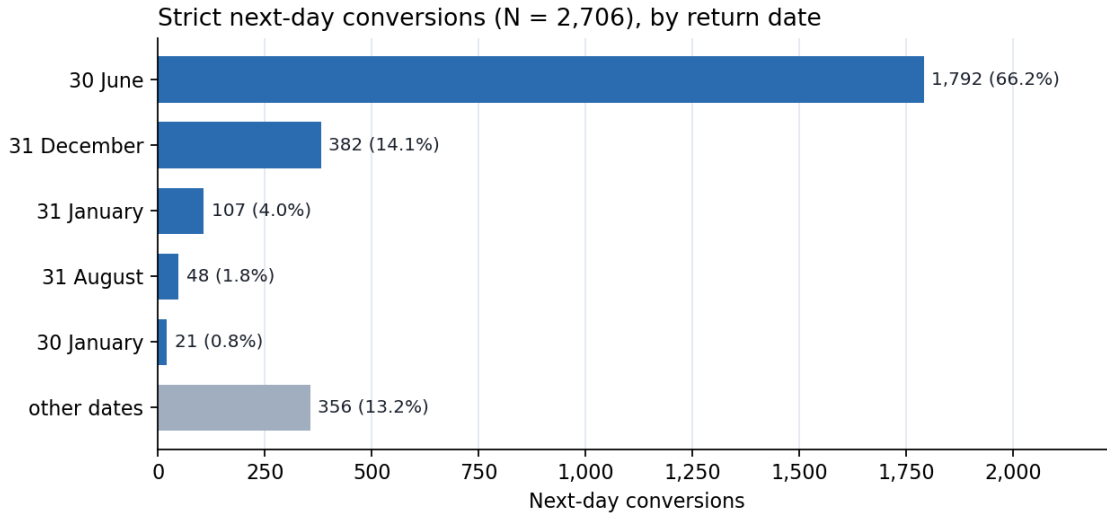


Figure 3: Return-date distribution among the 2,706 strict sequences whose permanent move occurs exactly one day after the loan return.

The largest return dates are:

Return date	Next-day conversions	Share
30 June	1,792	66.2%
31 December	382	14.1%
31 January	107	4.0%
31 August	48	1.8%
30 January	21	0.8%
Other dates	356	13.2%
Total	2,706	100.0%

In particular,

$$66.2\% + 14.1\% = 80.3\%$$

of the next-day conversions follow a return dated either 30 June or 31 December.

Thus, next-day conversions are strongly concentrated around standard season-ending dates. This is consistent with a permanent registration becoming effective immediately after the formal loan return.

However, the sequence alone does **not** establish whether the permanent move resulted from:

- a purchase option,
- a purchase obligation,
- another pre-agreed mechanism, or
- a newly negotiated permanent transfer.

Those mechanisms require separate contract research.

3 The British 31 May convention

A pure day-count rule treats different football calendars differently.

For many loans, the common pattern is:

30 June return $\rightarrow$ 1 July permanent transfer,

which appears as a one-day lag.

For British loans, a comparable sequence may instead appear as:

31 May return $\rightarrow$ 1 July permanent transfer,

which is recorded as a 31-day lag.

As a descriptive sensitivity, I therefore counted a case as “calendar-immediate” if either:

1. the permanent move occurs within one day, or
2. the return is on 31 May and the permanent move is on 1 July of the same year.

The result is:

Definition	Loans	Share of 4,068
Within 1 day	2,708	66.6%
Within 1 day, or 31 May → 1 July	2,769	68.1%
Additional cases from calendar rule	61	1.5 percentage points

The calendar convention therefore matters conceptually, but it does not materially change the overall result: roughly two-thirds of strict conversions are immediately adjacent in registration time.

4 When do loans end?

The previous sections condition on loans that eventually become permanent at the same borrower. I next looked at the calendar distribution of **all realised loan endings**.

There are:

$$N = 27,402$$

realised loan endings with an observed ending date.

Figure 4 shows both the most frequent month-day combinations and the cumulative coverage obtained by taking the most common dates first.

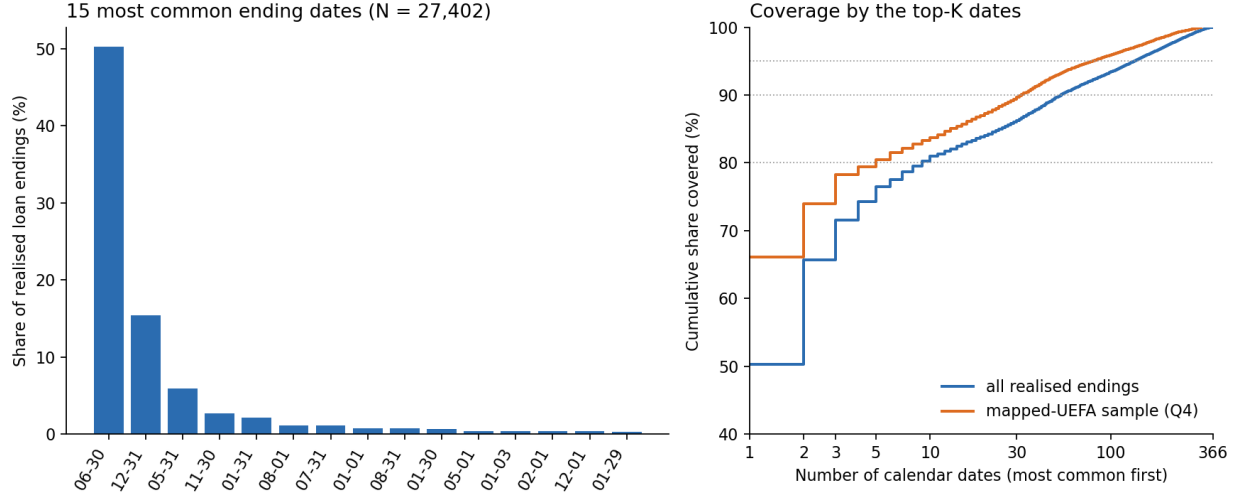


Figure 4: Left: the 15 most common calendar dates among all realised loan endings. Right: cumulative coverage as increasingly many recurring calendar dates are included, for all realised endings and for the mapped-UEFA sample used in the earlier analysis.

The ten most common dates are:

Rank	Ending date	Share of realised endings
1	30 June	50.3%
2	31 December	15.4%
3	31 May	5.9%
4	30 November	2.7%
5	31 January	2.1%
6	1 August	1.1%
7	31 July	1.1%
8	1 January	0.8%
9	31 August	0.7%
10	30 January	0.7%

The concentration is substantial at the top, but falls quickly after the largest few dates:

Dates included	Cumulative share
Most common date	50.3%
Top 3 dates	71.6%
Top 5 dates	76.5%
Dates required for 80%	9
Dates required for 90%	53
Dates required for 95%	135

Therefore, a short list of five universal dates does **not** capture 95% of realised loan endings. The top five capture only 76.5%.

5 Mapped-European comparison

The earlier analysis of European loans restricted the sample to cases where both the lending and borrowing club could be deterministically mapped to UEFA.

For the main realised European sample:

$$N = 10,924.$$

The comparison with all realised endings is:

Statistic	All realised endings	Mapped-UEFA sample
N	27,402	10,924
30 June	50.3%	66.1%
31 December	15.4%	4.3%
31 May	5.9%	7.9%
Top 3 dates	71.6%	78.3%
Top 5 dates	76.5%	80.5%
Dates required for 80%	9	5
Dates required for 90%	53	32
Dates required for 95%	135	79

The European sample is therefore more concentrated on a relatively small set of ending dates, particularly 30 June. Even there, however, a short universal list still does not reach 90–95% coverage.

6 Country heterogeneity

The calendar distribution is not the same across countries.

Among the **30 lender countries** with enough realised endings for the country-level comparison, **8 have a most common ending date other than 30 June**.

Examples include:

- Scotland, where 31 May is especially important;
- Brazil, where 31 December is prominent;
- the United States, where 30 November is important;
- Japan, where 31 January is prominent.

The global top three ending dates therefore describe some countries much better than others.

This suggests that if the objective is to classify loan endings as “standard” or “off-cycle”, a country-specific calendar definition may be preferable to a single global list.

Importantly, an ending outside a standard date should not automatically be called an early termination. It may instead reflect a different national calendar, a transfer-window date, a COVID-era schedule change, or some other registration pattern.

7 What happens after the other loan returns?

The strict $A \rightarrow B \rightarrow A \rightarrow B$ sequence is only one type of post-loan movement.

Figure 5 compares the timing of several broader follow-on categories.

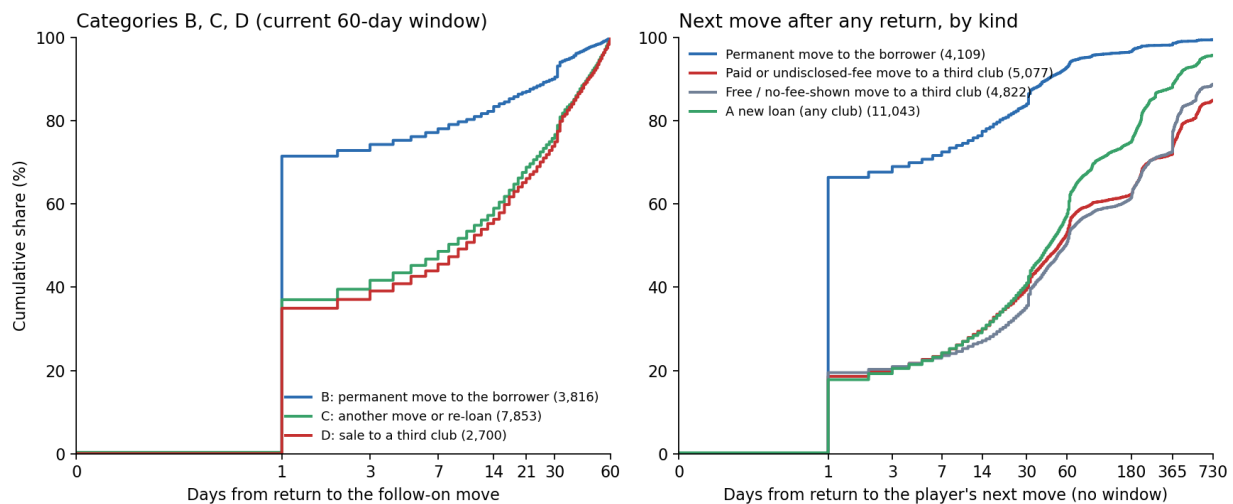


Figure 5: Left: timing of the current production follow-on categories inside the existing 60-day window. Right: timing of the player’s next substantive move after any loan return, with no maximum lag. The two panels therefore intentionally use different populations.

Left panel: current production categories

Under the current 60-day follow-on rules:

Category	Loans
Permanent move back to borrower	3,816
Another move or re-loan	7,853
Sale to a third club	2,700

These categories are defined using the existing production windows, so their lag distributions are truncated by construction.

Right panel: next substantive move at any distance

Ignoring the production window and simply looking for the next substantive move after a return gives:

Next move	Cases
Permanent move to the borrower	4,109
Paid or undisclosed-fee move to a third club	5,077
Free / no-fee-shown move to a third club	4,822
A new loan	11,043

Permanent moves back to the borrower are much more concentrated immediately after the return than the other types of subsequent movement.

This reinforces the interpretation that the next-day $A \rightarrow B \rightarrow A \rightarrow B$ pattern is a distinct registration pattern rather than simply reflecting the fact that all players tend to move again quickly.

8 What does this imply for the earlier 48.6% figure?

The previous analysis reported:

$$14,438/29,699 = 48.6\%$$

of loans as having a “non-ordinary surrounding registration sequence.”

That statistic is still useful as a descriptive production measure, but this week’s analysis shows that it is sensitive to the follow-on windows used to construct it.

The 14,438 cases consist of:

Component	Loans
Permanent move back to borrower within current rule	3,816
Other move or re-loan	7,853
Sale to third club	2,700
Loan ended through another movement, no return row	51
Fee-bearing return with no follow-on move	18
Total	14,438

Of these cases:

- **6,573** have their follow-on move within one day;

- **3,863** count as non-ordinary only because the follow-on occurs between 22 and 60 days after the return.

If one common follow-on window were applied to every type of move, the non-ordinary share would be approximately:

Follow-on window	Share classified non-ordinary
1 day	22.4%
60 days	51.9%
No maximum lag	84.5%

As the window becomes longer, the measure increasingly approaches the share of players who eventually move again.

Therefore, the earlier 48.6% should **not** be interpreted as the share of loans with unusual contractual clauses. It is partly a consequence of the chosen registration-sequence window.

9 Detailed day-by-day view

For completeness, Figure 6 shows the strict 4,068 sequences day by day.

The y-axis is intentionally clipped so that the much smaller day-by-day variation in the tail remains visible. The day-1 bar is therefore annotated rather than drawn to its full height, and all cases beyond day 120 are grouped into one final bar.

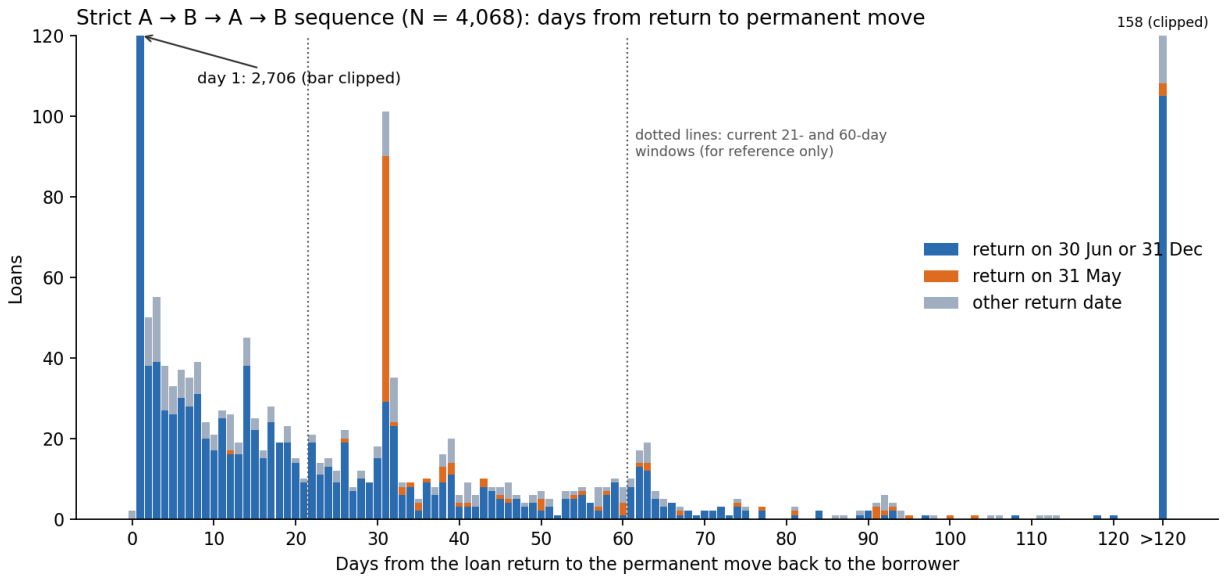


Figure 6: Day-by-day lag distribution for the 4,068 strict sequences. Day 1 contains 2,706 cases and is clipped to preserve visibility of the tail. The far-right bar aggregates all 158 cases with lags greater than 120 days.

This view also shows several smaller calendar-driven spikes, especially around 31 days and approximately 60–63 days. These are consistent with differences between season-ending dates and later registration or transfer-window dates.

10 Possible definitions going forward

The analysis suggests several possible definitions of an “immediate conversion.”

Option 1: next-day conversion

Define immediate as a permanent move occurring on the same day or the day after the return.

This covers:

$$2,708/4,068 = 66.6\%.$$

This is the clearest definition suggested directly by the distribution.

Option 2: calendar-aware immediate conversion

Define immediate as either:

- within one day, or
- a 31 May return followed by a 1 July permanent move in the same year.

This covers:

$$2,769/4,068 = 68.1\%.$$

The advantage is that it treats the British 31 May convention more similarly to a 30 June → 1 July transition.

Option 3: wider numerical window

Possible wider thresholds include:

Window	Share captured
3 days	69.1%
7 days	72.7%
21 days	81.0%
60 days	93.0%

These broader definitions capture progressively more observations, but no additional sharp empirical breakpoint appears after day 1.

The appropriate rule therefore depends on what economic concept we want the variable to represent.

11 Next step: connect the loan episodes to playing time

The next step is to link the transfer episodes to player playing-time data.

The existing tables make this technically feasible.

Available join

The loan table can be linked to the appearances table using:

- player ID,
- club ID,
- game / appearance date.

The appearances table contains `minutes_played` directly, with no missing values on actual appearance rows.

For the **28,255 loans since 2012/13**:

- the borrowing club has appearance coverage during the loan for **9,209 loans**, or **32.6%**;
- conditional on the borrowing club being observed, the player appears at least once for the borrower in **80.7%** of cases.

Coverage limitation

Appearance data are available primarily for the 14 long-covered European leagues, together with parts of their domestic cups and UEFA Champions League and Europa League matches.

The leagues added to the upstream dataset in 2024 do not have comparable appearance coverage.

Therefore:

no appearance rows outside the covered competitions should be interpreted as missing data, not as zero playing time.

Unused substitutes also require care: they appear in the lineup data but do not have a minutes value in the same way as players who entered the pitch.

Possible playing-time outcomes

Before building the final variable, the relevant outcome should be chosen. Possible measures include:

1. minutes played at the borrower during the loan;
2. share of available minutes played during the loan;
3. minutes at the lender after the player returns;
4. minutes in the season following the loan;
5. some combination of loan-period and post-loan playing time.

12 Questions for discussion

The main decisions for the next stage are:

1. What should count as an immediate conversion?

The strongest empirical break is next-day: 66.6% of strict conversions occur within one day.

Should the variable use:

- a next-day definition,
- a calendar-aware next-registration definition, or
- a wider fixed window?

2. Should standard loan-ending dates be country-specific?

The three most common global dates explain 71.6% of realised endings, but calendar conventions differ substantially by country. A single global definition can therefore classify normal endings in some countries as unusual.

3. What playing-time outcome is most useful?

For the next analysis, should the primary outcome be:

- playing time at the borrower during the loan,
- playing time after returning to the lender,
- subsequent-season playing time,
- or multiple outcomes?

Summary

The main findings from this week's analysis are:

1. There are **4,068 exact** $A \rightarrow B \rightarrow A \rightarrow B$ loan-return-permanent sequences.

2. **2,706 (66.5%)** of the permanent transfers occur exactly one day after the return; including the two same-day cases gives **66.6% within one day**.
3. There is a large discontinuity between day 1 and day 2 (**2,706 versus 50 cases**). No comparable break appears at 21 or 60 days.
4. **80.3% of next-day conversions** follow a return dated either 30 June or 31 December.
5. A calendar-aware allowance for 31 May $\rightarrow$ 1 July raises the immediate-conversion share only from **66.6% to 68.1%**.
6. Among all **27,402 realised loan endings**, 30 June, 31 December and 31 May account for **71.6%**. Five dates account for only **76.5%**; reaching 90% requires 53 distinct calendar dates.
7. Ending-date conventions vary materially across countries, suggesting that a country-specific definition of a standard ending may be preferable to one universal global list.
8. The earlier 48.6% “non-ordinary sequence” statistic is strongly dependent on the follow-on window and should not be interpreted as the share of loans with unusual contract clauses.
9. Playing-time data can be joined next, although reliable borrower-side appearance coverage exists for only about one-third of loans since 2012/13.